

HAT-P-32b

R-1143

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_201227 · created 2026-07-31T00:16:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459210.6709 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.012
Transit depth (Rp/Rs)^2	2.97 +/- 0.4 [%]
Orbital Inclination (inc)	88.53 +/- 1.1
Airmass coefficient 1 (a1)	1.0368 +/- 0.0095
Airmass coefficient 2 (a2)	-0.029 +/- 0.026
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	#3 - [473, 280]
Optimal Aperture	3.72
Optimal Annulus	14.25
Transit Duration (day)	0.0869 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.172 $\pm$ 0.012 vs 0.1489 (deviation 1.45 σ)
Duration fit vs published	0.0869 d vs 0.1304 d (deviation 23.44 σ)
Mid-time O-C	8.2 min
Depth SNR	10.8
Residual scatter	0.91 %

Light curve

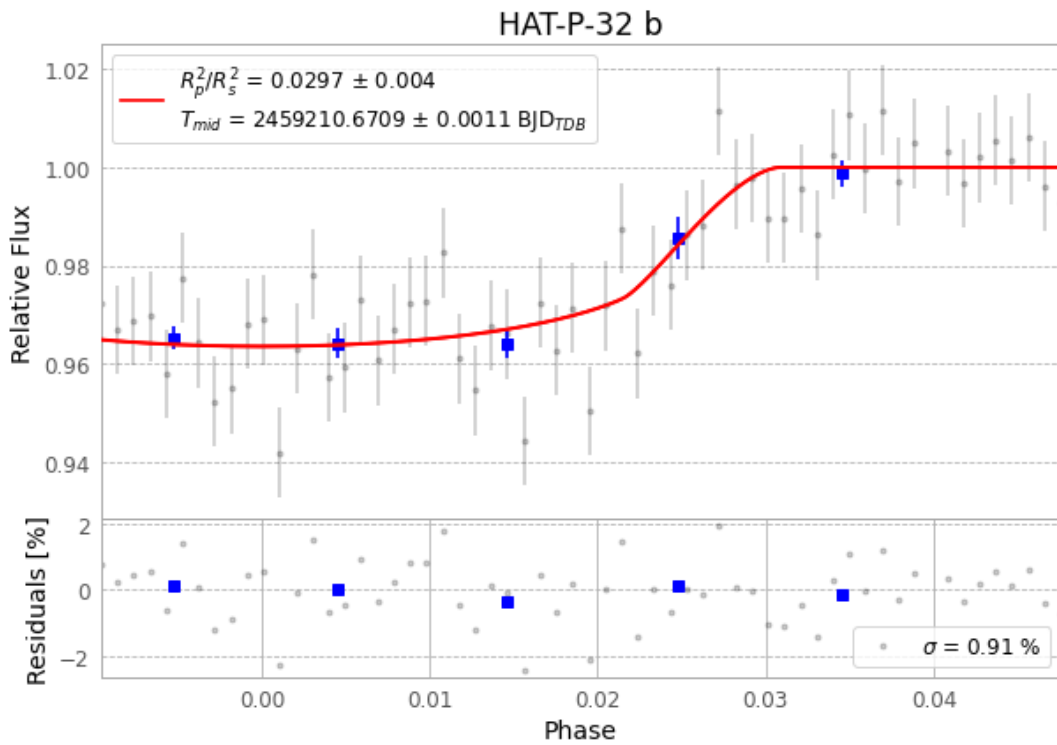


Figure 1 — FinalLightCurve_HAT-P-32 b_2020-12-26.png (SHA-256 a73a86954291348e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [490, 356], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d0aa88da548c54c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), HATP-32201227033012.FITS → HATP-32201227062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-201227.log (67cbb976afbe...), FinalLightCurve_HAT-P-32 b_2020-12-26.png (a73a86954291...), FinalLightCurve_HAT-P-32 b_2020-12-26.csv (fc32f48a0f66...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:16Z.